

Research Interests

Humanoid Robot Learning; Whole-Body Skill Learning; Reinforcement Learning; Motion Imitation; Motion Prior Learning; Sim-to-Real Transfer; Contact-Rich Robot Control

Education

Nanjing University of Posts and Telecommunications (NJUPT)

B.Eng. Candidate in Artificial Intelligence | Expected Graduation: 2028

GPA: 4.08/5.0 | Ranked 3/133 in Artificial Intelligence

Research Experience

Learning Dynamic Skills for Humanoid Robots via Reinforcement Learning

2026–Present

T1 Humanoid Robot Kicking and Locomotion

- Implemented and modified reinforcement-learning environments for humanoid locomotion and contact-rich kicking tasks using MuJoCo/MJLab/Warp simulation workflows.
- Implemented and evaluated task components including action and observation spaces, ball reset logic, contact detection, reward functions, termination conditions, and curriculum learning procedures.
- Conducted experiments to analyze policy behavior through contact quality, ball velocity, target direction error, post-kick recovery, and training-stability metrics.
- Investigating Conditional AMP and motion-prior-based kicking policies for skill generalization across different ball positions and kicking directions.

Conditional AMP for Generalizable Humanoid Kicking

2026

RL Research Project

- Investigated distance-conditioned Conditional AMP for humanoid kicking using condition vectors over target distance and kick type with curriculum learning.
- Studying how expert-policy condition alignment and motion priors influence training stability and cross-condition skill generalization.

Motion Prior Learning for Humanoid Skill Acquisition

2026

Research Exploration

- Studying AMP-style motion priors, human motion data, encoder-decoder policies, and latent representations for humanoid skill learning.
- Exploring methods for balancing task rewards and motion-prior rewards to achieve natural and effective robot behaviors.

Robot Hand-Eye Calibration with 3D Sensing and Geometric Constraints

2026

Research Project | Manuscript in Preparation | Nanjing University of Posts and Telecommunications

- Conducting research on robot hand-eye calibration with a completed patent application and ongoing manuscript preparation.
- Investigating multi-stage calibration methods based on orthogonal geometric constraints, point-cloud filtering, and optimization.
- Conducted calibration experiments with ABB industrial robots and 3D sensing systems for evaluation.

Technical Skills

Programming: Python, C++, Git/GitHub, Linux

Machine Learning / RL: PyTorch, Reinforcement Learning, PPO, reward design, curriculum learning

Robot Learning: Humanoid motion control, contact-rich skill learning, motion imitation, AMP-style motion priors

Simulation: MuJoCo, MJLab, Warp

Robotics Software: ABB RAPID Programming, Robot Calibration, 3D Sensing, Point-Cloud Processing

Awards

- Third Place, RoboCup China Open 3D Simulation League, 2025.
- Third Place, Jiangsu Provincial Robot Competition 3D Simulation Group, 2025.
- Third Prize, National Undergraduate Electronic Design Contest, Jiangsu Province, 2024.